STAT410: Intro to Probability Theory

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1. Probability: The Mathematical Study of Uncertainty

Definition 1.1

A **probability space** $(\Omega, \mathcal{F}, P)$ is composed of

(i) The **sample space** Ω. It contains all possible outcomes of the "experiment".

(ii) The **event space** $\mathcal{F}$ consisting of **events**. An event is a collection of outcomes.

(iii) The **probability measure** $P : \mathcal{F} \to \mathbb{R}$.

Example 1.2. *Toss a fair coin. This has the sample space* $\Omega := \{H, T\}$*, the event space* $\mathcal{F} = \mathbb{P}(\Omega)$*, and the probability measure* $P : \mathcal{F} \to \mathbb{R}$ *given by*

$$P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2, \quad P(\emptyset) = 0, \quad P(\{H, T\}) = 1$$

Example 1.3. *Roll a fair dice. This has the sample space* $\Omega = \{1, \ldots, 6\}$*, the event space* $\mathcal{F} = \mathbb{P}(\Omega)$*. Note that* $\mathcal{F}$ *is too big to succinctly write what the probability measure* P *does. More on this later.*

Example 1.4. *Pick a number uniformly at random from* $[0, 1]$*. Here,* $\Omega = [0, 1]$*. We pretend that* $\mathcal{F} = \mathbb{P}([0, 1])$*, but in reality* $\mathcal{F} \subsetneq \mathbb{P}(\Omega)$*. Again, we can't explicitly write what* P *does.*

Definition 1.5

We define the following axioms for the event space $\mathcal{F}$:

1. $\emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
3. If $\{A_i\}$ is a sequence of events in $\mathcal{F}$, then $\bigcup A_i \in \mathcal{F}$.

This is the definition of a **σ-algebra**.

Remark 1.6. *If* $A_1, \ldots, A_n \in \mathcal{F}$*, then* $\bigcup_{i=1}^n A_i \in \mathcal{F}$.

Proof. Let $A_i = \emptyset$ for all $i > n$. Then, we know $\bigcup_{i=1}^n A_i = \bigcup_{i=1}^\infty A_i \in \mathcal{F}$ □

Proposition 1.7

If $\{A_i\}$ is a sequence of events in $\mathcal{F}$, then $\bigcap A_i \in \mathcal{F}$.

Proof.

$$\bigcap_{i=1}^\infty A_i = \left(\bigcup_{i=1}^\infty A_i^c \right)^c \in \mathcal{F}$$

by axiom 2, 3, and De Morgan's law. □

Definition 1.8

We define the following axioms for the probability measure P:

1. $P(\emptyset) = 0$ and $P(\Omega) = 1$.
2. For any $A \in \mathcal{F}$, $P(A) \geq 0$.

3. (Countable additivity) If $\{A_i\}$ is a sequence of pairwise-disjoint events, then
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Example 1.9. *Let $\Omega = \{1, \dots, 6\}$ be the sample space for rolling a fair dice, and $\mathcal{F} = \mathbb{P}(\Omega)$. Recall that $P(\Omega) = 1$, and moreover*
$$\Omega = \{1\} \sqcup \{2\} \sqcup \cdots \sqcup \{6\}.$$
Thus,
$$1 = P(\Omega) = \sum_{i=1}^{6} P(\{i\}).$$
By the assumption that the dice is fair, we know $P(\{i\}) = P(\{j\})$, hence $P(\{i\}) = 1/6$ for all $i \in \{1, \dots, 6\}$.

Example 1.10. *Suppose $\Omega = [0, 1]$, and for any subinterval $I \subset [0, 1]$, $P(I) =$ length of I. What is the probability you see a rational number? An irrational number?*

Note that for any $x \in [0, 1]$, $P(\{x\}) = 0$ since $\{x\}$ can be considered a closed interval with length 0. Thus, since $\mathbb{Q}$ is countable, it follows that
$$P(\mathbb{Q} \cap [0, 1]) = 0, \quad P((\mathbb{R} \setminus \mathbb{Q}) \cap [0, 1]) = 1$$

Proposition 1.11

$$P(B^c) = 1 - P(B)$$

2. Continuity of Probability Measures

Let $B_1 \supset B_2 \supset \cdots$ be a decreasing sequence of sets. Then, we write
$$\lim_{n\to\infty} B_n = B = \bigcap_{i=1}^{\infty} B_i,$$
which is the greatest lower bound, or the infimum.

Theorem 2.1

The probability measure P commutes with limits. That is, given an ascending sequence $C_1 \subset C_2 \subset \cdots$,
$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\lim_{n\to\infty} C_n\right) = \lim_{n\to\infty} P(C_n)$$

Proof. Let $D_1 = C_1$, $D_2 = C_2 - C_1$, and generally $D_{i+1} := C_{i+1} - C_i$. Then, observe
$$\bigcup_{i=1}^{\infty} C_i = \bigcup_{i=1}^{\infty} D_i,$$
hence
$$P\left(\bigcup C_i\right) = P\left(\bigcup D_i\right).$$

Moreover, since the D_i's are pairwise disjoint, one shows that

$$P(D_{i+1}) = P(C_{i+1}) - P(C_i),$$

hence

$$\sum_{i=1}^{n} P(D_i) = P(C_n).$$

Thus,

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} C_i\right) = P\left(\bigcup_{i=1}^{\infty} D_i\right) &= \sum_{i=1}^{\infty} P(D_i) \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} P(D_i) = \lim_{n\to\infty} P(C_n) \end{aligned}$$

as desired. □

3. Inclusion-Exclusion Principle

Inclusion-Exclusion Principle **3.1**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof. Observe

$$P(A \cup B) = P(A \cup (B - A)) = P(A) + P(B - A).$$

Moreover,

$$P(B) = P(A \cap B) + P(B - A) \implies P(B - A) = P(B) - P(A \cap B),$$

thus

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

□

Corollary **3.2**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Proposition **3.3**

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)$$

Proposition **3.4**

If $A \subset B$, then $P(A) \leq P(B)$.

4. Intro to Random Variables

Definition 4.1

A **random variable** is a (measurable) function $X : \Omega \to \mathbb{R}$.

Definition 4.2

The **cumulative distribution function**, abbreviated **cdf**, is

$$F_X(a) := P(X \leq a) = P(X \in [-\infty, a]).$$

Now, suppose X takes countably many values, i.e. $\{X(\omega) \mid \omega \in \Omega\}$ is countable. A **probability mass function**, abbreviated **pmt**, is a function p_X such that there's sequence of real numbers $\{x_n\}$ with $p_X(x_n) = P(X = x_n) > 0$ for all n, and $p_X(x) = P(X = x) = 0$ for all other x.

Example 4.3. *Any experiment with finitely many outcomes X is called a discrete random variable. Then, $P_X(x)$ is the probability mass function of X. The set where $P_X(x) > 0$ is called the **support**.*

For any set B, we have

$$P(x \in B) = \sum_{x \in \text{supp}(X) \cap B} p_X(x).$$

Definition 4.4

A random variable X is said to be **(absolutely) continuous** if there exists a function $f_X(x)$ such that $f_X(x) \geq 0$, and for any interval $I = [a, b]$,

$$P(x \in I) = \int_a^b f_X(x)dx$$

5. Conditional Probability

Question 5.1. *Does knowing A has taken place affect our chances of seeing B?*

Definition 5.2

Let A, B be events. Then, the probability of B given A is denoted $P(B|A)$.

Theorem 5.3

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Proof. We can give a frequentist proof with

$$P(B|A) = \lim_{n \to \infty} \frac{\frac{\#A,B \text{ both take place}}{n}}{\frac{\#A \text{ takes place}}{n}} = \frac{P(A \cap B)}{P(A)}.$$

□

Example 5.4. *An experiment consists of tossing a fair coin, followed by rolling a fair D6 if you see heads, and rolling a fair D4 if you see tails. What is the probability that you see heads and two?*

If $A =$ coin is heads and $B =$ dice is two, then our probability is

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.$$

Theorem 5.5

$P'(X) := P(X|A)$ is a probability measure. However, $P''(X) := P(B|X)$ is not a probability measure.

***Law of Total Probability* 5.6**

Let $A_1, \ldots, A_n$ partition Ω. and $P(A_i) > 0$ for all i. Then,

$$P(B) = \sum_{i \in I} P(B|A_i)P(A_i)$$

This also holds for countable partitions.

***Bayes' Theorem* 5.7**

Suppose $A_1, \ldots, A_n$ partition Ω, and $P(A_i) > 0$. Suppose further that B is an event with $P(B) > 0$. Then, for any $j \in \{1, \ldots, n\}$,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)},$$

as desired.

Proof. By Theorem 5.3,

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)},$$

and by the law of total probability

$$\frac{P(A_j \cap B)}{P(B)} = \frac{P(A_j \cap B)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)} = \frac{P(B|A_j) \cdot P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

□

Example 5.8. *The prevalence of a particular disease is 1%. A rapid test for this disease has a 2% false positive rate, and a 3% false negative rate. If the test comes up positive, how likely is it that you have the disease?*

Solution. Let D be the event that you have the disease, and T be the event that you tested positive. Then,

$$P(D) = 0.01,$$
$$P(T|D^c) = 0.02,$$
$$P(T^c|D) = 0.03.$$

By Bayes' Theorem,

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)} = \frac{(1-0.03)\cdot 0.01}{(1-0.03)\cdot 0.01 + 0.02\cdot(1-0.01)} = 0.3288$$

□

6. Independence

Definition 6.1

Events A and B are said to be **independent** of $P(A \cap B) = P(A)P(B)$.

Theorem 6.2

Let A, B be events with $P(A), P(B) > 0$. If they are independent, they cannot be disjoint.

Proof. $P(A \cap B) = P(A)P(B) > 0$. □

Theorem 6.3

For any events $A_1, \ldots, A_n$ with $P(A_i) > 0$,

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P\left(A_n | \bigcap_{i=1}^{n} A_i\right) P\left(A_{n-1} | \bigcap_{i=1}^{n-2} A_i\right) \cdots P(A_1).$$

Example 6.4. *An experiment consists of tossing a fair coin until you see heads for the first time, at which point the experiment terminates. Let A_i be the event you see tails on the i'th toss. Are the A_i's independent? What is the probability we see heads for the first time on the n'th toss?*

No, the A_i's are not independent. Next, let B_n be the event that heads is seen on the n'th toss. Then,

$$B_n = A_1 \cap A_2 \cap \cdots \cap A_{n-1} \cap A_n^c.$$

Therefore,

$$P(B_n) = P\left(A_n^c | \bigcap_{i=1}^{n-1} A_i\right) \cdot \prod_{k=1}^{n-1} P\left(A_k | \bigcap_{i=1}^{k-1} A_i\right) = (1/2)^n.$$

Exercise 1. *If A, B are independent, show that A, B^c are independent.*

7. Random Variables

Definition 7.1

For a random variable X, and a (measurable) $A \subset \mathbb{R}$, we define the distribution D_X to be

$$D_X(A) = P(X \in A)$$

Now, given a random variable $X : \Omega \to \mathbb{R}$, we have

$$D_X(A) = P(X \in A) = P\left(\{\omega \mid X(\omega) \in \mathbb{R}\}\right),$$

and

$$F_X(a) = P(X \leq a) = P(X \in (-\infty, a]).$$

Clearly, D_X determins F_X, but in fact we'll see that F_X determines D_X.

Example 7.2. *Let X be the outcome of rolling a fair die. Then,*

$$F_X(a) = \begin{cases} 0 & a < 1 \\ 1/6 & 1 \le a < 2 \\ 2/6 & 2 \le a < 3 \\ \vdots & \\ 5/6 & 5 \le a < 6 \\ 1 & 6 \le a. \end{cases}$$

In other words,

$$F_X(a) = \begin{cases} 0 & a < 1 \\ 1 & a \ge 6 \\ \frac{\lfloor a \rfloor}{6} & 1 \le a < 6. \end{cases}$$

Proposition 7.3

The function $F_X : \mathbb{R} \to \mathbb{R}$ satisfies the following properties:

1. $\lim_{X \to \infty} F_X(a) = 1$ and $\lim_{X \to -\infty} F_X(a) = 0$
2. F_X is monotonically increasing
3. F_X is right continuous, i.e.

$$\lim_{x \to a^+} F_X(x) = F_X(a)$$

Definition (*Discrete Random Variables*) 7.4

X is called **discrete** if there is a countable $A \subset \mathbb{R}$ such that $P(X \in A) = 1$.

Then, the function $a \mapsto P(X = a)$ is called the **probability mass function** (pmf), and the probabilites of interest are computed by taking a suitable sum of the pmf.

Definition (*Continuous Random Variables*) 7.5

X is called **(absolutely) continuous** if there is some function $f_X(x)$ such that $f_X(x) \ge 0$, and for any interval I with endpoints $-\infty \le a \le b \le \infty$,

$$P(X \in I) = \int_a^b f_X(x)dx$$

Note that we then have

$$F_X(a) = \int_{-\infty}^a f_X(x)dx$$

Definition (*Identical Distributions*) 7.6

Random variables X, Y are said to be **identically distributed** if $D_X = D_Y$, which holds if and only if $F_X = F_Y$.

A good way to think about identical distributions is with an example: how do you generate a random integer in $\{0,1\}$?

One way to do it is flipping a coin, taking 0 if the coin is tails, and 1 if the coin is heads. Another way to do it is generating a random real number in $[0,1]$, taking 0 if it's $< 1/2$, and 1 if it's $\geq 1/2$.

Let X be the coin method and Y be the real number method. Note that X and Y have completely different sample spaces:

$$X : \{H,T\} \to \mathbb{R} \quad X(\{H\}) = 1, X(\{T\}) = 0$$
$$Y : [0,1] \to \mathbb{R} \qquad Y(\omega) = \begin{cases} 0 & \omega < 0.5 \\ 1 & \omega \geq 0.5. \end{cases}$$

However, observe

$$F_X(a) = F_Y(a) = \begin{cases} 0 & a < 0 \\ 1/2 & 0 \leq a < 1 \\ 1 & a \geq 1. \end{cases}$$

Thus, X and Y are identically distributed, even though they're completely different (they have different sample spaces!).

Moreover, let $W = 1 - X$, i.e. $W = 1$ on tails and $W = 0$ on heads. Then, X and W are still identically distributed!

Remark 7.7. 1. *If F_X is piecewise constant, then X is a discrete random variable. In this case, $P(X = a)$ can be computed by $F(a) - \lim_{X \to a^-} F_X(a)$*

2. *If X is (absolutely) continuous, F_X is continuous*

3. *If F_X is differentiable with only finitely many exceptions, then $F_X'(x) = f_X(x)$ for all points where F_X is differentiable. At other points, we may define f_X arbitrarily.*

Example 7.8. *Suppose X is a continuous random variable with pdf $f_X(x) = 1/2$ for $x \in [-1,1]$.*

(i) Compute $P(X > 0)$

We know

$$P(X > 0) = \int_0^\infty f_X(x)dx = \int_0^1 \frac{1}{2}dx = 1/2$$

(ii) Compute $P(X > 1/2 | X > 0)$.

Using conditional probability, we see that

$$P(X > 1/2 | X > 0) = \frac{P(X > 1/2 \cap X > 0)}{P(X > 0)} = \frac{1/4}{1/2} = \frac{1}{2}.$$

(iii) Compute the pdf of X^2.

Observe that

$$P(X^2 \leq a) = P(-\sqrt{a} \leq X \leq \sqrt{a}) = \int_{-\sqrt{a}}^{\sqrt{a}} f_X(x)dx.$$

When $0 \leq a < 1$, this evaluates to $\sqrt{a}$. Thus, we see

$$F_{X^2}(a) = \begin{cases} 0 & a < 0 \\ \sqrt{a} & 0 \leq a < 1 \\ 1 & a \geq 1. \end{cases}$$

Note that this is the cdf of X^2. Differentiating, we get

$$f_{X^2}(a) = \begin{cases} 0 & a < 0 \\ \frac{1}{2\sqrt{a}} & 0 \leq a < 1 \\ 0 & a \geq 1. \end{cases}$$

Definition **7.9**

We say that a random variable X has a **uniform distribution** on $[a, b]$, denoted $X \sim u[a, b]$, if

$$f_X(x) = \frac{1}{b-a}$$

for $x \in a, b$.

Example 7.10. *Suppose $X \sim u[0, 1]$. Find the pdf of $Y := e^X$.*

Solution. If $a < 1$, then e^X doesn't attain a on $[0, 1]$, so $F_Y(a) = 0$ for $a < 1$. Similarly, when $a > e$, we know with certainty that $e^X < a$, so $F_Y(a) = 1$. Finally, if $a \in [1, e]$, then

$$F_Y(a) = P(Y \leq a) = P(e^X \leq a) = P(X \leq \log a) = P(0 \leq X \leq \log a) = \log a.$$

Thus,

$$F_Y(a) = \begin{cases} 0 & a < 1 \\ \log(a) & 1 \leq a \leq e \\ 1 & a > e, \end{cases}$$

and differentiating yields

$$f_Y(a) = \begin{cases} 0 & a \leq 1 \\ \frac{1}{a} & 1 < a < e \\ 0 & a \geq e. \end{cases}$$

□

8. Expected Values

Definition (*Expected Value (Discrete)*) **8.1**

Let X be a discrete random variable with support A. Then, the **expected value** of X, denoted $E[X]$, is

$$E[X] = \sum_{x \in A} x p_X(x) = \sum_{x \in A} x P(X = x)$$

Definition (*Expected Value (Continuous)*) **8.2**

Let X be a continuous random variable. Then, the **expected value** of X, denoted $E[X]$, is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Definition **8.3**

We say that the expected value $E[X]$ **exists** if $E[|X|] < \infty$.

Example 8.4. *Suppose $X \sim u[0,1]$. Compute $E[X]$ and $E[e^X]$.*

Solution. By definition,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x dx = \frac{1}{2}.$$

For $E[e^X]$, one method involves using the pdf we computed earlier, which gives

$$E[e^X] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_1^e \frac{y}{y} dy = e - 1.$$

However, we may use the following general result:

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx,$$

which gives

$$E[e^X] = \int_0^1 e^x dx = e - 1.$$

□

LOTUS **8.5**

Let X be a random variable with density function f_X. Then, for $g : \mathbb{R} \to \mathbb{R}$,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Linearity of Expectation **8.6**

If X, Y are random variables, and a is a real number, then $E[aX + Y] = aE[X] + E[Y]$.

Markov's Inequality **8.7**

Suppose $X \geq 0$ and $c \in \mathbb{R}^+$. Then

$$P(X \geq c) \leq E[X]/c,$$

Chebyshev's Inequality **8.8**

Let X be a random variable with finite mean μ and finite variance σ^2. For any $c \in \mathbb{R}^+$,

$$P\left(|X - \mu| \geq c\right) \leq \frac{\sigma^2}{c^2}$$

Proposition 8.9

If $E[X^k]$ is finite, then $E[X^n]$ is finite for all $n \in \{1, \dots, k\}$. In particular, for Chebyshev's inequality, it's sufficient to know that σ^2 is finite.

Proof. Observe

$$E\left[|X^m|\right] = \int_{-\infty}^{\infty} |x^m| f_X(x) dx = \int_{|x^m|>1} |x^m| f_X(x) dx + \int_{|x^m|\leq 1} |x^m| f_X(x) dx$$

by LOTUS (Theorem 8.5), for any m. Let A be the first term above when $m = k$, and B the second term when $m = k$. When $x^n > 1$, then $|x^n| f_X(x) \leq |x^k| f_X(x)$. When $|x^m| \leq 1$, then $|x^m| f_X(x) \leq f(x)$. Thus,

$$E[|X^n|] \leq \int_{|x^n|>1} |x^k| f_X(x) dx + \int_{-\infty}^{\infty} f_X(x) \leq A + 1 < \infty,$$

so $E[X^n]$ is finite. □

9. Special Distributions

In this section, we study some important distributions to know.

Discrete Distributions

9.1. Bernoulli Distribution

Definition 9.1

A **Bernoulli trial** is a random variable which has two outcomes: success (1) and failure (0).

Proposition 9.2

A bernoulli random variable X has a probability mass function

$$p_X(x) = \begin{cases} 1-p & x = 0 \\ p & x = 1 \end{cases}$$

where p is the probability of success. Moreover,

$$E[X] = p$$
$$V[X] = p(1-p)$$

Proof. We compute

$$E[X] = \sum x p_X(x) = 0 \cdot (1-p) + 1 \cdot p = p$$
$$V[X] = E[X^2] - E[X]^2 = p - p^2 = p(1-p)$$

□

Example 9.3. *Repeat a Bernoulli trial in an independent manner. Let X be a random variable that counts*

the number of trials until the first success. Then, we know X has the probability mass function

$$p_X(x) = p(1-p)^{x-1} \qquad x = 1, 2, \dots$$

Moreover,

$$E[X] = \sum_{n=1}^{\infty} n \cdot p(1-p)^{n-1} = \frac{1}{p}$$

To compute this sum, use the derivative trick. Remember the sums and derivatives commute when we're within the radius of convergence.

Question 9.4. *The probability of an independent event happening is p. What's the expected number of attempts does it take for the event to happen?*

Solution. Let X be the number of attempts to make the event happen, and we saw in the previous example that $X \sim \text{Geom}(1/10^7)$. Then, $E[X] = 10^7$. □

9.2. Binomial Distribution

We repeat a Bernoulli trial, with success chance p, n times, where n is fixed. Let X be the random variable counting the number of successes.

Let's compute $P(X = k)$. This involves k successes, which has probability p^k, and $n - k$ failures, with probability $(1-p)^k$. Now, we must count the number of ways in which we can achieve k successes. This is $\binom{n}{k}$, and so

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Moreover,

$$E[X] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}.$$

This sum can be computed with some neat tricks. However, we will eventually see that X can be written $X = X_1 + \cdots + X_n$ as the sum of n Bernoulli random variables, which are independent and identically distributed (**IID**). Note that $E[X_i] = p$, so $E[X] = np$.

9.3. Hypergeometric Distributions

Say we want to sample a portion of a population for some property. Do we sample with or without replacement? If we sample with replacement, we get a binomial distribution. However, it can be better to sample without replacement, in which case we get a hypergeometric distribution.

Definition 9.5

Let N, N_A, n be integers with $0 \le n \le N_A \le N$. A random variable X is said to follow a **hypergeometric distribution** with respect to the parameters (N, N_A, n) if X takes values in $\{1, \dots, n\}$, and

$$P(X = k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}.$$

This is interpreted as follows: there are $\binom{N}{n}$ ways to select n people out of your population of N. Then, if N_A of those people have the condition, there are $\binom{N_A}{k}$ ways to find k of them, and $\binom{N-N_A}{n-k}$ for the rest of them to not have it.

Example 9.6. *Suppose that a collection of 20 rare books contains 5 first editions, each costing* \$5000*, and 15 second editions, each costing* \$1000*. Four books are picked at random to be sold. What is the expected income?*

Solution. Let X be the number of first editions, so $4 - X$ is the number of second editions. Then, the revenue is

$$r(X) := 5000X + 1000(4 - X) = 4000X + 4000.$$

By linearity of expectation,

$$E[r(X)] = E[4000X + 4000] = 4000E[X] + 4000.$$

We're now left to compute $E[X]$. Note that X is sampling randomly from a population without replacement, so it's a hypergeometric distribution. The population size is $N = 20$, the number of first editions is $N_A = 5$, and the sample size is $n = 4$. Computing $E[X]$ is left as an exercise. □

9.4. Poisson Distribution

Definition 9.7

Fix $\lambda > 0$. A random variable X is said to have a **Poisson distribution** with parameter λ if $X \geq 0$ is integer valued, and has a probability mass function

$$p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}$$

for $k \in \{0, 1, 2, \dots\}$.

Remark 9.8. *If* $X \sim \text{Poisson}(\lambda)$*, then* $E[X] = V[X] = \lambda$.

Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Poisson}(np)$. Then, for any $A \subset \{0, 1, 2, \dots\}$, we have

$$|P(X \in A) - P(Y \in A)| \leq np^2$$

Example 9.9. *Assume that a random variable X counts the occurrence of rare events that are not "strongly dependent". Then, the distribution of X can be approximated with a* $\text{Poisson}(E[X])$ *distribution.*

Example 9.10. *Suppose that a large factory experiences, on average, three accidents each month. What is the probability that on a particular month there are two accidents?*

Solution. Since the factory is large, we can assume that three accidents per month is very small, so we may assume that the number of accidents is represented by $X \sim \text{Poisson}(3)$. Then,

$$P(X = 2) = e^{-3}\frac{3^2}{2!} \approx 0.224$$

□

Continuous Distributions

There are three types of continuous distributions: uniform, exponential, and normal.

Uniform distributions try to compute the idea that the regions with equal "length" have the same probability of having something "drawn" from them.

9.5. Uniform Distribution

Definition 9.11

A random variable X is said to have a **uniform distribution**, denoted $X \sim \text{Unif}[a, b]$ for $a < b$ if

$$f_X(x) = \frac{1}{b-a}, \quad x \in [a, b]$$

9.6. Exponential Distribution

Definition 9.12

Let $0 < \lambda < \infty$. The **exponential distribution** with rate λ is given by $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$. If X is a random variable with this distribution, we write $X \sim \exp(\lambda)$.

Note that the exponential distribution is related to the geometric distribution.

Proposition 9.13

Let X be a continuous random variable with an exponential distribution with rate λ. Then

$$E[X] = 1/\lambda,$$
$$V[X] = 1/\lambda^2.$$

Proof. We compute

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx = \lim_{h\to\infty} \left[-xe^{-\lambda x}\Big|_0^h + \int_0^h e^{-\lambda x} dx \right] = \frac{1}{\lambda}.$$

The variance computation is similar. □

Proposition 9.14

Let X be a continuous random variable with an exponential distribution with rate λ. Then,

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

Example 9.15. *A library has 5 computers for public use. The average lifetime of a computer is 6 years, and follows an exponential distribution. What is the probability that at least one computer is working after 7 years? You may assume the lifetimes are independent.*

Solution. Let T_i be the lifetime of computer $i \in \{1, \dots, 5\}$. Then, $T_i \sim \exp(1/6)$, and

$$P(T_1 > 7 \vee \cdots \vee T_5 > 7) = 1 - P(T_1 \leq 7, \cdots, T_5 \leq 7) = 1 - \prod_{i=1}^{5} P(T_i \leq 7) = 1 - P(T \leq 7)^5$$

where $T \sim \exp(1/6)$. We computed in Proposition 9.14 that $P(T \leq 7) = 1 - e^{-7/6}$. Thus,

$$\cdots = 1 - (1 - e^{-7/6})^5$$

□

9.7. Normal Distributions

Definition **9.16**

A continuous random variable X is said to have a **normal distribution** with parameters μ, σ^2 (mean and variance) if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

for $-\infty < x < \infty$.

Central Limit Theorem (Philosophy) **9.17**

For large samples, the sample average follows a normal distribution.

Example 9.18. *The height of males in a particular island can be modeled by a normal distribution with mean* 170 *and standard deviation* 5*. What is the probability that a randomly picked male has a height that balls between 160 and 180?*

Solution. Let X be the height of a randomly picked male, so $X \sim N(170, 25)$. Then

$$P(160 < X < 180) = P\left(\frac{160-170}{5} < \frac{X-\mu}{\sigma} < \frac{180-170}{5}\right) = P(-2 < Z < 2)$$

where $Z = (X - \mu)/\sigma$. Note that $Z \sim N(0, 1)$, and so we've translated this problem into one about the standard normal.

Assume we have a table (called a Z-table) which gives values of $\Phi(z) = P(Z \leq z) = F_Z(z)$. Then

$$P(a < Z < b) = \Phi(b) - \Phi(a).$$

Figure 1 is an example of a Z-table. Using it, we get

$$\Phi(2) - \Phi(-2) \approx 0.9772 - 0.228 \approx 0.95.$$

□

-2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
−1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
−1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
−1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
−1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
−1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
−1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
−1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08692	0.08534	0.08379	0.08226
−1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
−1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
−1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
−0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
−0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
−0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
−0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
−0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
−0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
−0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
−0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
−0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
−0.0	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
z	−0.00	−0.01	−0.02	−0.03	−0.04	−0.05	−0.06	−0.07	−0.08	−0.09

(a) Z-table of negatives

z	+ 0.00	+ 0.01	+ 0.02	+ 0.03	+ 0.04	+ 0.05	+ 0.06	+ 0.07	+ 0.08	+ 0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56360	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169

(b) Z-table of positives

Figure 1: Z-table

10. Moment Generating Functions

Definition **10.1**

The ***k*-th moment** of a random variable X is defined as $E[X^k]$.

Remark 10.2. *Recall that if $E[X^k] < \infty$, then $E[X^m] < \infty$ for $m \in \{1, \dots, k\}$, but the converse does not hold.*

Definition **10.3**

A **moment generation function** of a random variable X is defined as

$$\mu_X(t) = E[e^{tX}].$$

We say that the moment generating function exists if $\mu_X(t)$ is finite on some open interval $(-\varepsilon, \varepsilon)$.

Theorem **10.4**

Let X be a random variable. Then,

$$\mu_X^{(n)}(t) = E[X^n e^{tX}],$$
$$\mu_X^{(n)}(0) = E[X^n].$$

Example 10.5. *Let $X \sim \exp(\lambda)$. Then*

$$\begin{aligned}\mu_X(t) = E[e^{tx}] &= \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \\ &= \lambda \int_0^{\infty} e^{tx} e^{-\lambda x} dx\end{aligned}$$

$$= \lambda \lim_{h\to\infty} \int_0^h e^{(t-\lambda)x} dx.$$

If $t \geq \lambda$, this diverges. Otherwise,

$$\mu_X(t) = \frac{\lambda}{t-\lambda} \lim_{h\to\infty} \left[e^{(t-\lambda)x}\right]_0^h = \frac{\lambda}{\lambda - t}.$$

Thus

$$\mu_X(t) = \begin{cases} \frac{\lambda}{\lambda - t} & t < \lambda, \\ \infty & t \geq \lambda. \end{cases}$$

Now computing the variance is easy. Namely,

$$\begin{aligned} V[X] = E[X^2] - E[X]^2 &= \mu_X''(0) - \left[\mu_X'(0)\right]^2 \\ &= \frac{2\lambda}{(\lambda - t)^3}\bigg|_{t=0} - \left(\frac{\lambda}{(\lambda - t)^2}\right)^2\bigg|_{t=0} \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \end{aligned}$$

This aligns with Proposition 9.13.

Example 10.6. *Let $Z \sim N(0,1)$. Then*

$$\begin{aligned} \mu_Z(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} \cdot e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{2tz - z^2}{2}\right) dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(t^2/2\right) \exp\left(-(t-z)^2/2\right) dz \\ &= \frac{\exp(t^2/2)}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{-(t-z)^2}{2}\right) dz \\ &= \frac{\exp(t^2/2)}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{-z^2}{2}\right) dz \\ &= \frac{\exp(t^2/2)}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = \exp(t^2/2). \end{aligned}$$

11. Random Vectors

Definition **11.1**

Let $X_1, \ldots, X_n : \Omega \to \mathbb{R}$ be a collection of random variables. We define the **random vector** $\vec{X} : \Omega \to \mathbb{R}^n$ to be

$$\vec{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix},$$

also written $\vec{X} = (X_1, \ldots, X_n)$. The probability distribution of X is called the **joint distribution**.

Proposition 11.2

If X, Y are random variables, then

$$p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y|x).$$

Proof. By definition

$$\begin{aligned} p_{X,Y}(x,y) &= P(X=x, Y=y) \\ &= P(X=x) \cdot P(Y=y|X=x) \\ &= p_X(x) \cdot p_{Y|X}(y|x), \end{aligned}$$

as desired. □

Example 11.3. *We flip a coin, and if heads we roll a D6, and if tails we roll a D4. Then*

$$\Omega = \{(H,1), \ldots, (H,6), (T,1), \ldots, (T,4)\}.$$

We write

$$X(a,b) = \begin{cases} 0 & a = T \\ 1 & a = H \end{cases}$$

$$Y(a,b) = b$$

Compute the joint probability mass function $p_{X,Y}(x,y)$/

Solution. By Proposition 11.2,

$$p_{X,Y}(x,y) = p_X(x) \cdot p_{Y|X}(y|x).$$

Note that $X \sim \text{Ber}(1/2)$, and $p_X(x) = 1/2$ when $x \in \{0,1\}$. Moreover

$$\begin{aligned} p_{Y|X}(y|0) &= 1/4, \quad y \in \{1, \ldots, 4\}, \\ p_{Y|X}(y|1) &= 1/6, \quad y \in \{1, \ldots, 6\}. \end{aligned}$$

Hence

$$p_{Y|X}(y|x) = \begin{cases} 1/4 & x = 0, y = 1, \ldots, 4 \\ 1/6 & x = 1, y = 1, \ldots, 6 \\ 0 & \text{otherwise.} \end{cases}$$

We conclude that

$$p_{X,Y}(x,y) = \begin{cases} 1/8 & x = 0, y = 1, \ldots, 4 \\ 1/12 & x = 0, y = 1, \ldots, 6 \\ 0 & \text{otherwise.} \end{cases}$$

□

Question 11.4. *What is the probability that the die roll in Example 11.3 is prime?*

Solution. We are computing

$$p_{X,Y}(0,2) + p_{X,Y}(0,3) + p_{X,Y}(1,2) + p_{X,Y}(1,3) + p_{X,Y}(1,5).$$

This evaluates to

$$\frac{1}{8}+\frac{1}{8}+\frac{1}{12}+\frac{1}{12}+\frac{1}{12}=\frac{1}{2}.$$

However, we kay take another approach: if we compute $p_Y(y)$, then all we need is $p_Y(2)+p_Y(3)+p_Y(5)$. Well, by the law of total probability,

$$p_Y(y)=p(Y=y)=p(Y=y|X=0)+p(Y=y|X=1).$$

We compute that

$$p_Y(y)=\begin{cases}5/24 & y=1,2,3,4\\ 1/12 & y=5,6,\end{cases}$$

and so $p_Y(2)+p_Y(3)+p_Y(5)=5/24+5/24+1/12=1/2$. □

Proposition 11.5

For random variables X, Y,

$$p_Y(y)=\sum_x p_{X,Y}(x,y).$$

Definition 11.6

The random variables $X_1,\dots,X_n$ are said to be **independent** if

$$p_{X_{i_1},\dots,X_{i_k}}(x_{i_1},\dots,x_{i_k})=\prod_{j=1}^{k}p_{X_{i_j}(x_{i_j})}$$

for all $(i_1,\dots,i_k)\subset(1,\dots,n)$.